# The Effect of Weather Severity on Students' Willingness to Attend In-Person Classes

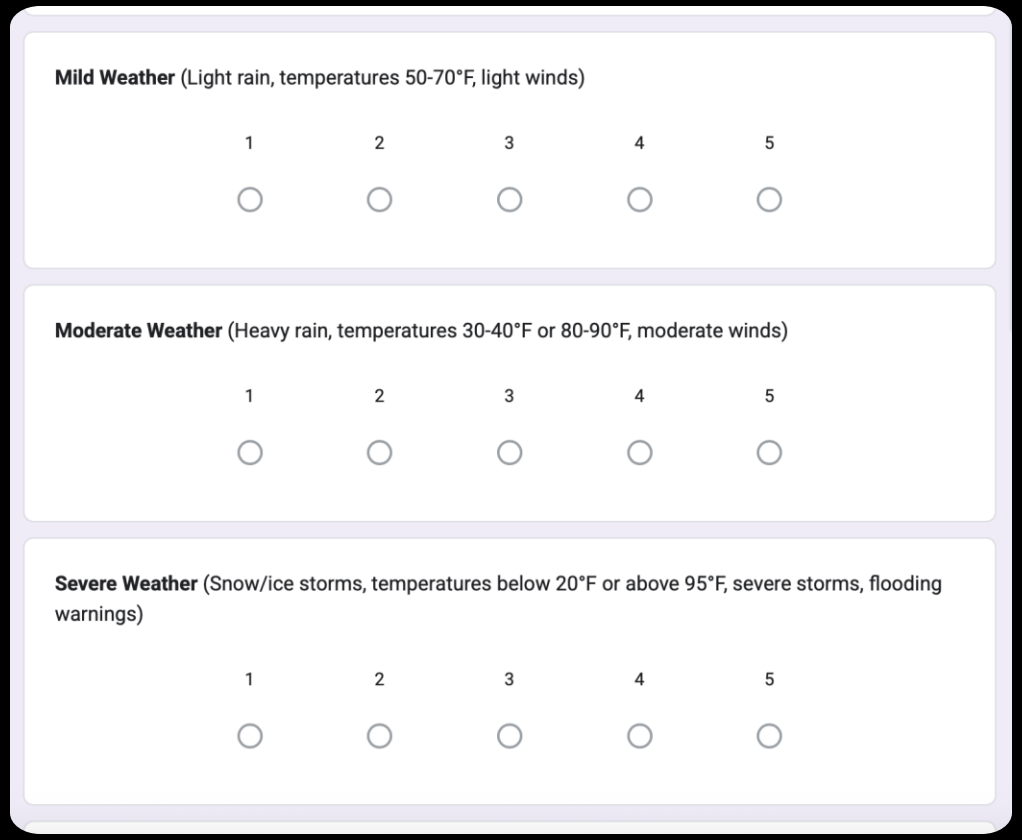
A Statistical Study

Srikala Sandu, Rifat Rahman Khan

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# Motivation

Research Question

- Do weather conditions (mild, moderate, severe) have an effect on students' willingness to attend in-person classes?

This was a repeated-measures survey; we binarized willingness ratings (1–2 = low, 3–5 = high) and used Cochran's Q test to compare responses across weather conditions.

Weather Condition Scenarios

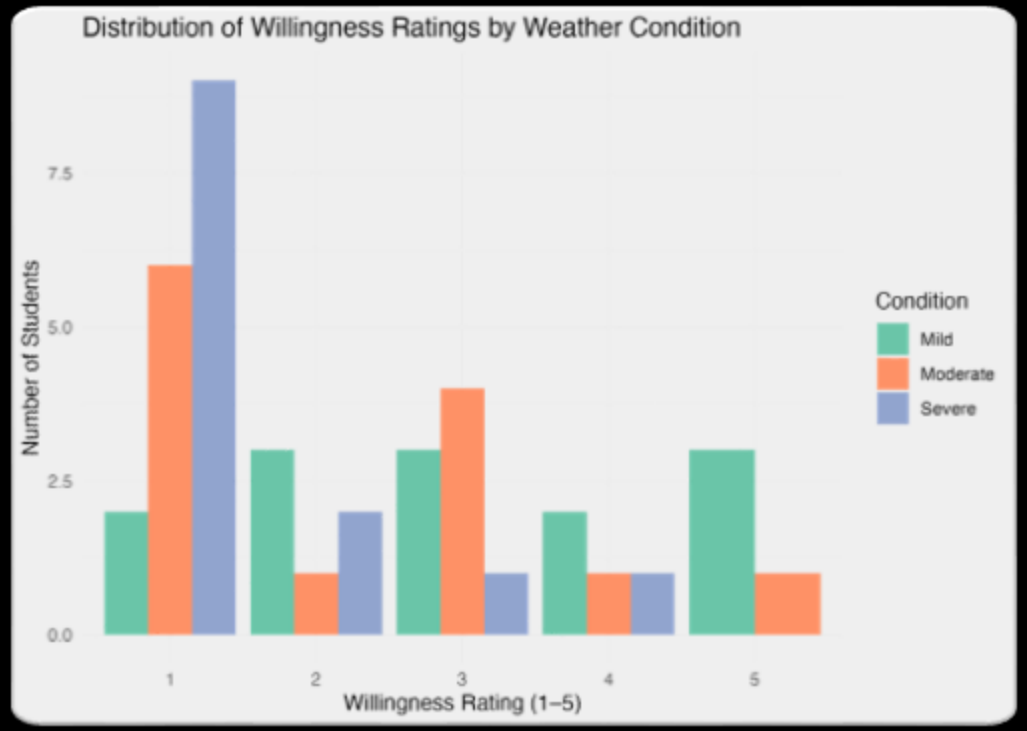
For each weather scenario below, rate your willingness to attend in-person classes on a scale of 1-5:

1 to 2 represent that students prefer online and 3-5 represents students prefer offline

# Data

- Participants: 13 respondents
- 11 Graduate students
- 2 PhD students
- Tool: Google Forms
- Recruitment: Shared via social media groups and personal connections
- Limitations: Small sample size, mostly from one institution, voluntary response

# Results




- We initially considered a Chi-squared test of independence, but it was not appropriate due to repeated measurements from the same students.
- We therefore merged ratings 1–2 and 3–5 to create a binary outcome and applied Cochran's Q test, which is suitable for repeated-measures categorical data.
- The Cochran's Q test was statistically significant ($Q(2) = 9.33$, $p = 0.0094$); using a bootstrap-based power estimation method showed high power (~0.94), so we reject the null hypothesis that willingness is the same across weather conditions.